DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture Week 7: Assignment

Random Walks · Scaled Limits · Brownian Motion

Instructions

- This assignment covers Lectures 19–21 and the Week 7 Lab. It carries **100 marks** total.
- Show all working for numerical questions – correct answers with no working receive partial credit only.
- Questions marked **[Python]** require a short script; submit your code alongside your written answers.
- You may use standard normal tables or software for probability calculations; state which you used.
- Submit a single PDF plus any `.py`/`.ipynb` files, via the course submission portal.

Part A – Concepts (15 marks)

A1. **(4 marks)** State the $\sqrt{n}$ law for a simple symmetric random walk and explain, without formulas, why the typical displacement grows like $\sqrt{n}$ rather than n.

A2. **(4 marks)** State the four defining properties of standard Brownian motion. Which one is responsible for the $\sqrt{\text{time}}$ scaling?

A3. **(4 marks)** Explain what Donsker's Theorem adds beyond the ordinary Central Limit Theorem, and give one concrete financial example where the stronger statement is genuinely needed.

A4. **(3 marks)** True or false, with justification: "Brownian paths are so irregular that they must contain jumps."

Part B – Random Walks (20 marks)

B1. **(6 marks)** For the simple symmetric random walk, compute $\mathbb{P}(S_6 = 2)$ and $\mathbb{P}(S_6 = 0)$ exactly, showing your combinatorial reasoning.

B2. **(5 marks)** A stock has annual volatility $\sigma = 30\%$. Using $\sqrt{\text{time}}$ scaling, compute its volatility over one month, one week (52 weeks/year), and one trading day (252 days/year).

B3. **(5 marks)** An *asymmetric* walk has $\mathbb{P}(X_i = +1) = 0.55$ and $\mathbb{P}(X_i = -1) = 0.45$. Compute $\mathbb{E}[X_i]$, $\text{Var}(X_i)$, and hence $\mathbb{E}[S_{100}]$ and $\text{Var}(S_{100})$. Is S_n a martingale? If not, is it a sub- or supermartingale?

B4. **(4 marks)** Prove directly from the definition that $S_n^2 - n$ is a martingale for the *symmetric* walk, showing each step of the conditional expectation calculation.

Part C – Scaled Limits (15 marks)

C1. **(6 marks)** Consider rescaling a random walk by $N^{-\alpha}$. Compute $\mathrm{Var}(S_N/N^{\alpha})$ as a function of α, and use it to explain precisely why $\alpha = \frac{1}{2}$ is the only value giving a non-degenerate limit. State what happens for $\alpha = 0.3$ and $\alpha = 0.8$.

C2. **(4 marks)** For the rescaled walk $W^{(N)}(t)$ on $[0, 1]$ with $N = 625$ steps, compute Δt and the step size, and verify explicitly that the step size equals $\sqrt{\Delta t}$.

C3. **(5 marks)** A random walk has i.i.d. steps taking values $+3$ and -3 with equal probability. Compute $\mathrm{Var}(S_n)$. If you rescale by $1/\sqrt{N}$, what is the variance of the limiting process at time t? How does this relate to the "universality" remark in Lecture 20?

Part D – Brownian Motion (25 marks)

D1. **(5 marks)** For standard Brownian motion, state the distribution of W_9 and compute $\mathbb{P}(W_9 > 4.5)$.

D2. **(7 marks)** For $s = 3$ and $t = 7$: (a) state the distribution of $W_7 - W_3$; (b) compute $\mathrm{Cov}(W_3, W_7)$, showing your reasoning; (c) compute the correlation between W_3 and W_7.

D3. **(6 marks)** Prove that $W_t^2 - t$ is a martingale, showing every step. (Hint: write $W_t = W_s + (W_t - W_s)$ and expand the square.)

D4. **(4 marks)** State the quadratic variation of standard Brownian motion over $[0, 5]$. Explain in your own words why $\sum(\Delta W)^2$ converges while $\sum|\Delta W|$ diverges, referencing the typical increment size.

D5. **(3 marks)** Using the self-similarity relation $W_{ct} \overset{d}{=} \sqrt{c}\, W_t$, verify that the standard deviation of W_9 computed this way (with $c = 9$, $t = 1$) agrees with your answer to D1.

Part E – Computational (20 marks)

E1. **(10 marks) [Python]** Using your Lab code, verify the martingale property $\mathbb{E}[W_T \mid W_s] = W_s$ at *three different* conditioning values: $W_s \in \{-0.8, 0.0, +1.2\}$, with $T = 1$, $s = 0.5$. Report the number of paths in each bin, the estimated conditional expectation, and the theoretical value. Comment on whether the accuracy differs across the three cases and, if so, explain why.

E2. **(10 marks) [Python]** Investigate how the quadratic variation estimate converges. For $T = 1$ and $n \in \{10, 10^2, 10^3, 10^4, 10^5, 10^6\}$, simulate *50 independent paths* at each n and report the mean and standard deviation of $\sum(\Delta W)^2$ across those 50 paths. Plot the standard deviation against n on log–log axes. Comment in 2–3 sentences on the rate at which the estimate's spread shrinks.

Part F – Integrative Question (5 marks)

F1. **(5 marks)** In Week 6 you observed the N-step binomial option price converging to a fixed number as N grew, and were told the limit was "the Black–Scholes price." In a short paragraph (100–150 words), explain what Week 7 has now revealed about *why* that convergence occurs. Your answer should reference the CRR parameterization, the logarithm connecting binomial prices to random walks, and Donsker's theorem.

End of Assignment. Solutions will be released after the submission deadline.